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Social network group decision making: Characterization, taxonomy, challenges and future directions from an AI and LLMs perspective

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ABSTRACT

In the past decade, social network group decision making (SNGDM) has experienced significant advancements. This breakthrough is largely attributed to the rise of social networks, which provides crucial data support for SNGDM. As a result, it has emerged as a rapidly developing research field within decision sciences, attracting extensive attention and research over the past ten years. SNGDM events involve complex decision making processes with multiple interconnected stakeholders, where the evaluation of alternatives is influenced by network relationships. Since this research has evolved from group decision making (GDM) scenarios, there is currently no clear definition for SNGDM problems. This article aims to address this gap by first providing a clear definition of the SNGDM framework. It describes basic procedures, advantages, and challenges, serving as a foundational portrait of the SNGDM framework. Furthermore, this article offers a macro description of the literature on SNGDM over the past decade based on bibliometric analysis. Solving SNGDM problems effectively is challenging and requires careful consideration of the impact of social networks among decision-makers and the facilitation of consensus between different participants. Therefore, we propose a classification and overview of key elements for SNGDM models based on the existing literature: trust models, internal structure, and consensus mechanism for SNGDM. This article identifies the research challenges in SNGDM and points out the future research directions from two dimensions: first, the key SNGDM methodologies and second, the opportunities from artificial intelligence technology, in particular, combining large language models and multimodal fusion technologies. This look will be analyzed from a double perspective, both from the decision problem and from the technology views.

1. Introduction

With the advancement of social media technology such as Weibo, WeChat, Facebook, and other platforms, it is easier for people to share and access information. By studying the flow of information and interactions among people on these platforms, huge amounts of data can be collected. These data help us understand how information or opinions spread among people and the impact they have within groups [1]. This kind of analysis is very important for studying people's behavior patterns and group dynamics. For example, during the pandemic, by analyzing information about vaccines on social media, it can understand the public's views and acceptance levels towards vaccines. This information is extremely valuable for governments and policymakers, as it can help to make better decisions, such as how to promote vaccine uptake and how to address public concerns [2]. Therefore, online users can participate in the decision making process through discussions, voting, negotiations, and other means on social networks, fully utilizing the diversity of group opinions and integrating collective wisdom to achieve better decision outcomes [3].

Social network is of significant importance for understanding group behavior, improving collective intelligence, and effectively solving group decision making issues. To do that, Wu and Chiclana [4] systemically proposed the social network group decision making (SNGDM),

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1566-2535/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

which becomes a specific research area within the discipline of decision sciences [5]. Particularly, Ji et al. [6] proposed an overlapping community-driven SNGDM that the IEEE intelligence computational society classified as a research frontier in 2023. In the past decade, an increasing number of articles on SNGDM have been published. These articles extend classical group decision making (GDM) problems by introducing social network relationships to theoretically innovate group decision making models [4–10]. In fact, SNGDM is not just a combination of social networks and GDM but an interconnected and coupled process. However, current studies in SNGDM lack a clear definition that describes its defining aspects/characteristics. An important feature is the presence of relationship networks with varying strengths within the SNGDM. Then, the mechanisms for the formation and evolution of social networks will affect the group decision making process and result in greater complexity. Another important feature is that trust will produce many different mechanisms for group consensus, such as trust feedback, trust incentive, trust game, etc. Therefore, trust relationships linking social networks should receive more attention [11,12].

The integration of social networks into traditional GDM paradigms introduces a multitude of complexities, particularly due to the intricate network of relationships characterized by varying strengths within SNGDM. These complexities manifest themselves in several key areas:

(a) SNGDM often involves a large number of decision-makers (DMs) [13]. These numerous DMs may belong to different interest groups or factions and can effectively facilitate consensus building and the progression of decision making by clustering them based on their opinion similarity or social network relationships. With the development of information technology, it is urgent to use numerous tools for auxiliary processing in effective SNGDM methods,[1] such as high-dimensional clustering, artificial intelligence (AI), large language models (LLM), and social network analysis (SNA).
(b) The goal of the SNGDM events is to gain more support from stakeholders and reach a consensus.[2] In order to avoid dissatisfaction among individual decision-makers and groups, it is crucial in SNGDM to measure the level of consensus among DMs, identify inconsistent DMs and improve the group's consensus level based on feedback approaches.
(c) It often requires mining the motivation behind the formation of social networks to manage SNGDM events, that is, whether the formation of social network relationships is based on trust in opinions or trust in emotions [14]. And it helps better guide decision making groups to reach consensus and improve the efficiency of the group decision making process by mining the motivations for social network formation.

Taking into account the characteristics and challenges frequently encountered in the above mentioned SNGDM problems, this study aims to address the following three aspects:

(1) To define a standardized characterization of the SNGDM framework, it focuses on the procedural differences in the SNGDM process. Provides a definition and characterization of SNGDM and summarizes the strengths and weaknesses of the SNGDM framework compared to classical group decision making frameworks to elaborate on the subject.
(2) To present the current hot topics in SNGDM research in the form of a taxonomy and overview, this article will outline the key elements (processes and methods) considered in SNGDM models in the literature. These include: (i) trust modeling (e.g., trust representation, trust propagation, and trust evolution); (ii) internal structures of social networks (e.g., group consensus measurement, group power structures, and social network community structures); and (iii) consensus reaching mechanisms (e.g., feedback mechanisms, opinion dynamics, and behavior management). The authors will introduce how these key procedures can help improve the efficiency of the decision making process for SNGDM problems.
(3) To discuss the challenges faced by SNGDM research and future directions in three dimensions: (i) how to overcome current limitations in SNGDM methods; (ii) the potential innovative applications of SNGDM in real-world scenarios; and (iii) the use of AI in general and LLM in particular together with fusion technologies for SNA. It is emphasized that LLMs is expected to play a particularly important role in being integrated with SNGDM models and systems in future research. This look will be analyzed from a double perspective, both from the decision problem and from the technology views.

The main contributions of this article are twofold. Firstly, it systematically and comprehensively analyzes and reviews the definition, characteristics, and taxonomy of SNGDM based on existing research. Secondly, it conducts a forward-looking analysis by examining future technological trends and offering an in-depth analysis of the challenges and potential of SNGDM research, particularly from the perspectives of AI and LLMs. The structure of this article is as follows. Section 2 introduces the definition and characteristics of the SNGDM framework, presents a bibliometric analysis of the SNGDM literature to motivate further contributions, and provides a preliminary classification of current SNGDM models based on trust representation and propagation methods. Section 3 presents a taxonomy of key elements found in SNGDM models in the literature, considering trust models, network structures and consensus mechanisms. Subsequently, we discuss the unresolved issues and future research directions from the perspectives of SNGDM methodologies, in Section 4. In Section 5, it is important to consider the use of AI technologies focused on LLM with the double look of SNGDM problems and technologies. Finally, the lessons learned and conclusions are drawn in Section 6.

2. SNGDM framework: characterization and bibliometric analysis

2.1. Characterization of SNGDM framework

In light of the continuous development of SNGDM research and its increasingly abundant real-world application scenarios, this section provides a formal definition of the SNGDM framework. Section 2.1.1 describes the formal concepts and characteristics of SNGDM. Considering the increasingly rich application scenarios, Section 2.1.2 discusses the processes of SNGDM, summarizing its advantages and challenges.

2.1.1. Definition and characterization of SNGDM framework

In the past ten years, SNGDM events have attracted increasing attention from researchers [15]. SNGDM refers to the utilization of collective wisdom and collaborative decision making methods within social networks to solve problems and make decisions, where group members can participate in the decision making process through discussions, voting, consultations, and other means, fully leveraging the diversity and collective wisdom within the group to achieve better decision making outcomes. In this subsection, the authors present a clear definition of the characteristics of a SNGDM event.[3] This characterization will be used to classify SNGDM scenarios.

In our daily lives, society faces numerous problems that require making a decision. For instance, urban renewal, elections of village

[1] https://doi.org/10.1109/EAIS58494.2024.10570029.
[2] https://doi.org/10.1016/j.ins.2024.121292.
[3] https://doi.org/10.1007/s10462-022-10361-8.

chiefs, and compensation plans for water pollution treatment. Most of these decisions involve more than one decision-maker, and there are different relationships among the decision-makers, such as clan blood relations, classmates, friends, etc. For example, in the election of a village chief, every villager needs to vote, and the person with the most votes can become the village chief. And each candidate may have clan blood relations with different villagers. In fact, the nature of a SNGDM event is not complex; it essentially involves GDM problems under different social network structures. However, as social network relationships and structures evolve dynamically, SNGDM events can exhibit a myriad of characteristics [10]. Therefore, to see beyond the phenomena to the essence, the authors provide a definition for SNGDM framework, as follows:

Definition 1 (*SNGDM Framework*)**.** There is a decision making event, if there are no fewer than three DMs evaluating different decision alternatives, and there exists a relationship among the DMs composed of nodes, edges, and the weights of the edges, which influences the information interaction among the DMs and decision making result, thus, this decision making event is referred to as social network group decision making.

Based on Definition 1, this complex process of an SNGDM framework can be summarized in terms of the following characteristics.

(1) Firstly, SNGDM is essentially a group decision making problem. It requires no fewer than three DMs to participate together and evaluate different decision alternatives. Furthermore, there must be a social network relationship among the DMs, that is, a network relationship composed of the DMs, the relationships between them, and the strengths of those relationships.
(2) Secondly, this social network relationship must have an impact on the decision making process and outcomes, the primary mechanism involving influencing the information exchange between the DMs, thus affecting the decision outcomes. In this process, trust relationships are often involved, specifically including (i) trust representation, (ii) trust propagation, (iii) trust evolution, and (iv) the analysis of network structures.
(3) Finally, in the group decision making process, the structure of the social network and trust in the social network will play a key role in reaching group consensus. The research of social network group decision making mainly includes: community overlap, subgroup clustering, and subgroup power structure, etc. The research on the effect of trust on consensus mainly includes: trust-based feedback mechanism, trust-based behavior management, trust incentive model, etc. They usually interact to promote consensus in group decision making in a social network.

2.1.2. Basic procedure, advantages and challenges of SNGDM framework

Based on the definition and characteristics of SNGDM event, this article proposes a basic procedure, advantages and challenges faced by SNGDM problems, as shown in Fig. 1. First, it is important to clarify the decision making issue and define the decision attributes, objectives, and the scope of the participants involved. Secondly, group members are convened for discussion and decision making through social networking platforms or other means. Thirdly, they engage in discussions on social networks sharing their opinions, and information through text, images, videos, etc. During the fourth step, different viewpoints and reasons can be presented to engage in debates and negotiations among group members, where innovative thinking is stimulated, and new solutions are proposed. After reaching a certain consensus in the fifth step, voting and decision making can be conducted based on the majority opinion or achieving a certain level of consensus. Finally, the decision will be implemented, and its effectiveness and outcomes will be regularly evaluated for making adjustments and improvements.

Compared with traditional decision methods, the advantages of SNGDM are: (1) it can invite multiple group members from different backgrounds through social networks to participate in the process of discussions and decision making, resulting in more opinions and perspectives; (2) it will bring a decision making process of transparency and fairness, because group members can publicly display their views and opinions, and engage in debates and negotiations; (3) it avoids the time and space constraints of traditional meetings with the use of online and offline real-time communication and discussion through social networks, which allows more efficient decision making; (4) Discussions and decisions on social networks can be recorded and saved, facilitating easy review of past decision making experiences as well as their replication. However, while having advantages, SNGDM also presents many challenges. For example, group members may face the problem of information overload due to the richness and complexity of information on social networks, making it difficult to filter and evaluate effective information. Opinions among members may gradually become radicalized during the process of SNGDM, leading to extreme or biased decision outcomes. In addition, individual opinions and rights may be ignored or suppressed, resulting in personal interests not being protected.

Table 1
The composition of bibliometric entities.

Key words	Number of literature	Number after filtering
'Social network'& 'Consensus'	2720	466
'Social network'& 'GDM'	2722	687
'Trust' & 'Consensus'	2776	306
'Trust' & 'GDM'	2872	341
Total	11 090	1800
Total (removing duplicates)		917

2.2. Bibliometric analysis of SNGDM literature

In order to systematically review and analyze the findings of previous research studies on SNGDM problems, a literature retrieval and mining process has been adopted by the authors to identify relevant studies on the topic. Firstly, using the keywords 'social network' and 'consensus', 'social network' and 'GDM', 'trust' and 'consensus', as well as 'trust' and 'GDM', we conduct an advanced search in the Web of Science database. The specific results are shown in Table 1. A total of 11,090 articles are summarized from 1990 to 2024 and 1,800 articles are obtained after rigorous manual screening. The process of manual screening is very time-consuming, as the authors need to read the abstracts of every literature. A shortcut is based on the research field and the authors' institutions. For literature outside the fields of computer science, automation, and management, such as in medicine and fisheries, the authors can determine irrelevance based on the title and directly delete them. For articles from some agricultural and astronomical institutions, after reading their titles, most are found to be less relevant and are directly deleted. For those literatures that cannot be judged based on the title, the authors will read through their abstracts to determine whether to include them in the bibliometric list or not. By combining the four sets of search results and removing duplicates, a total of 917 references were acquired as of September 25, 2024 to form the main body of bibliometric study.

The research on SNGDM was firstly presented by Wu and Chiclana [4] in 2014. Fig. 2(a) shows the publication and citation situation of SNGDM research in the past decade. It can be observed that the research on SNGDM is still in the development process and has not yet reached its peak. In the future, more valuable results will be born in this field. Meanwhile, Fig. 2(b) reflects the average annual citation situation of SNGDM over the past decade. The papers published in the inaugural year of SNGDM [4], have the highest average citations and are significantly higher than those published later, which indicates that the earliest literature on SNGDM has higher academic value. In addition, for these publications, the published years and journals are

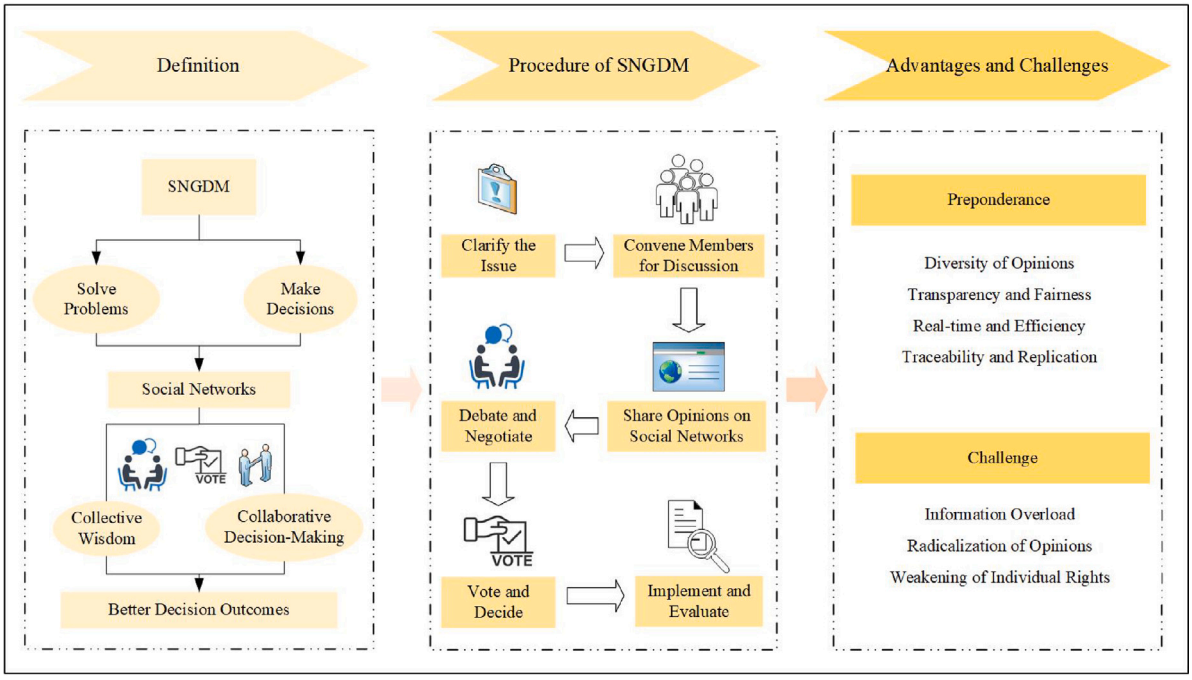


Fig. 1. The definition, procedure, advantages and challenges of SNGDM.

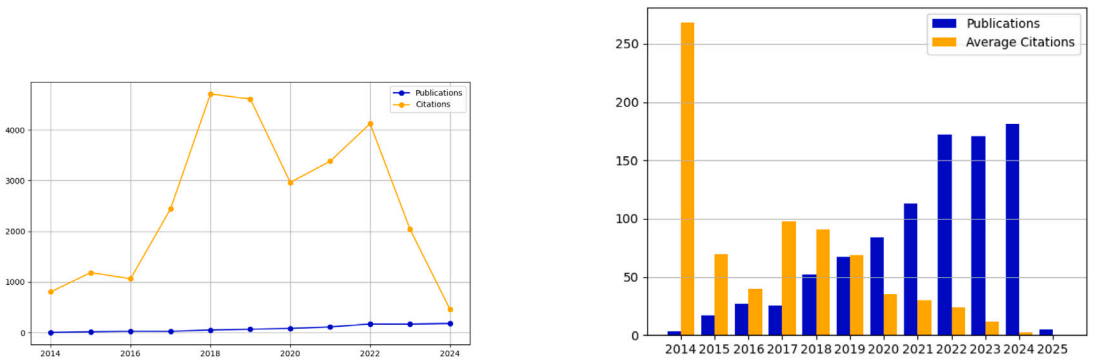


Fig. 2. The number of publications and citations for SNGDM from 2014 to 2024.

presented in Fig. 3,[4], it can be observed that in the past three years, *'Information Sciences'* and *'Information Fusion'* have been the two journals with the highest number of publications. With the development of LLMs technology, SNGDM will be significantly impacted in the future.[5]

Based on the analysis of authors from 917 literature sources, Fig. 4 presents the authors with collaborative relationships. The size of the edges connecting the scholars represents the number of joint publications in the SNGDM field. Different colors represent different clusters, that is, different academic schools. It can be seen that the authors of this review are very familiar with the topic and have been focusing on the SNGDM research field for the past decade, belonging to a significant academic school in this domain.

This paper explores the research hotspots in the field of SNGDM over the past decade based on keyword analysis. Fig. 5 presents the keywords of 917 articles, showing that current research on SNGDM is mainly centered on decision making methods, social network analysis techniques, consensus-reaching processes, etc. It represents just the surface of the issues in SNGDM. In fact, the deeper issues involve exploring trust behaviors and mechanisms between DMs. Although consensus reaching and network analysis are the most studied related topics about SNGDM, topics related to trust behavior also have gained special importance. In particular, issues related to trust modeling within SNGDM have received increasing attention and discussion, which will be elaborated on in Section 3.1.

3. Taxonomy of key elements in SNGDM literature

This section provides an overview of the models and methods involved in SNGDM from three aspects: trust modeling of SNGDM, internal structure of SNGDM, and consensus mechanism of SNGDM, which are illustrated in Fig. 6.

[4] In Fig. 3 the numbers in the circles represent the number of publications on each year in related journals, and circles of different colors correspond to the journals on the right..

[5] https://doi.org/10.48550/arXiv.2403.15587.



Fig. 3. The publications on each year and the related journals.

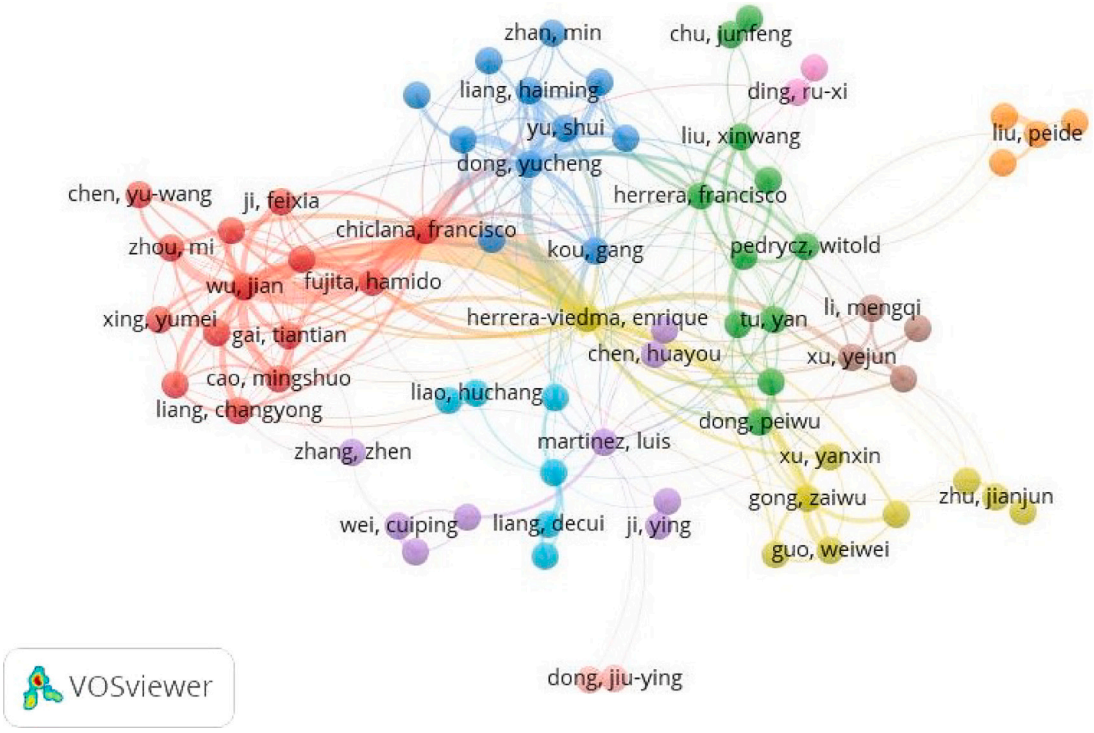


Fig. 4. The active authors in SNGDM research field from 2014 to 2024.

3.1. Trust modeling of SNGDM

In real SNGDM scenarios, the initial trust network constructed of decision-makers is usually incomplete, and trust propagation methods have been studied to evaluate the degree of indirect trust between decision-makers. Moreover, it is difficult to objectively reflect the dynamic update of trust relationships caused by the group interaction process with the constructed static trust relationship. Therefore, it is imperative to establish a social network trust evolution mechanism. Based on the results of trust modeling, it is necessary to further explore



Fig. 5. The keywords among the 917 publications about SNGDM.

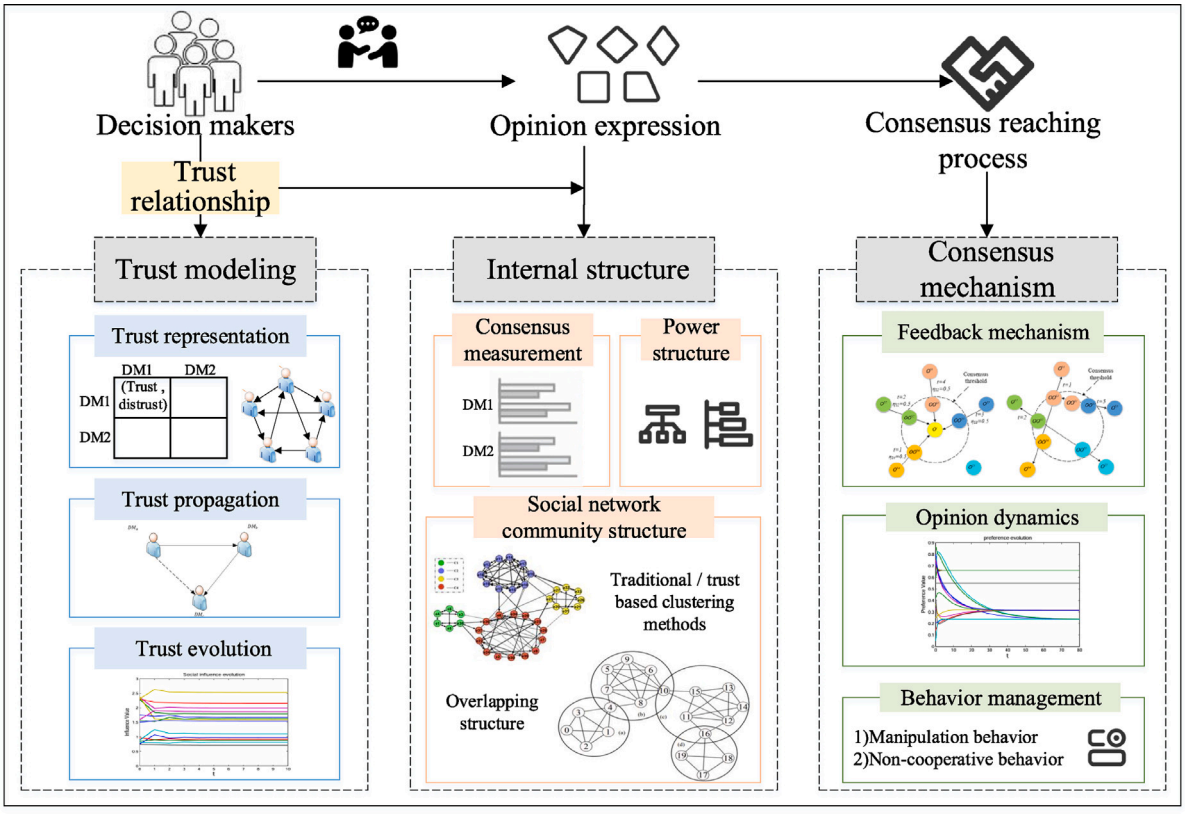


Fig. 6. Taxonomy of key elements in SNGDM.

the mechanism of social network overlapping nodes in the consensus interaction, which is crucial to achieving consensus in SNGDM.

The following provides a review of three aspects: trust propagation in social network, trust evolution in social network, and overlapping communities in social network.

3.1.1. Trust representation in social network

Trust is often manifested in many different forms on social networks, leading to a nebulous concept of trust. To facilitate investigation, trust behavior must be presented in a quantifiable manner using a numerical or categorical representation format.

Trust relationship representation methods fall mainly into the following four categories: (a) binary discrete values [16,17]; (b) multi-discrete values [18]; (c) continuous values [19]; and (d) fuzzy logic values [20,21] and so on. Binary discrete values refer to the most direct method for DMs to express trust in other DMs using a binary indicator: trust/distrust. Although binary discrete values are easily understood by DMs, in some cases trust and distrust may not be sufficient to represent the viewpoint of the party being trusted. Therefore, a new trust representation is introduced, that is, multiple discrete values such as 'very trust', 'trust', 'untrust', and 'very untrust', where DMs can evaluate other DMs more accurately. Continuous values are a common way to characterize trust, and many research results belong to this category. Meanwhile, some research uses fuzzy logic to represent trust due to the uncertainty of trust. Typical fuzzy logic-based trust representations include interval values, intuitionistic fuzzy numbers, trust score spaces, and other methods.

3.1.2. Trust propagation in social network

The primary function of the trust propagation model is to assist the trusting party in assessing the trust value between them and unknown trusted parties in the network, playing an extremely important role in constructing complete trust relationships in social networks. Some DMs may not be directly connected in social networks, so there is no direct trust relationship. Therefore, it is necessary to build trust relationships between these DMs without direct connection through trust propagation. The goal of a trust propagation system is to use known trust values to estimate unknown trust values between certain DMs. Generally speaking, trust propagation is characterized by the following four features: (i) Symmetry, (ii) Follow-through, (iii) Directness, (iv) Divergence. Let $DM = \{DM_a, DM_b, DM_c, DM_d\}$ be a set of decision-makers, for symmetry, if DM_a trusts DM_b, it can be inferred that DM_b also trusts DM_a to some extent, shown as Fig. 7(a). If DM_a and DM_b both trust DM_c, and DM_b also trusts DM_d, then it is possible for DM_a to develop trust in DM_d following the trust of DM_b, shown as Fig. 7(b), which is called by the follow-through of trust. While the directness of trust means that if DM_a trusts DM_b, and DM_b trusts DM_c, then it can be derived that DM_a also trusts DM_c, shown in Fig. 7(c). Regarding the divergence of trust, if DM_a trusts DM_b, and both DM_b and another DM_c trust DM_d, then the trust that DM_a has in DM_b may lead to DM_a also developing trust in DM_c, shown in Fig. 7(d).

Currently, the two main approaches to trust propagation include: (a) trust propagation operator and (b) trust aggregation method [22]. Usually, trust propagation operators mainly include: (1) multiplicative transitive operators [23,24], (2) selective transitivity [17], (3) matrix transitivity [25], (4) t-norm transitivity [26,27], (5) trust propagation based on social theory [28,29], (6) trust propagation methods based on machine learning [30,31] and so on. Although the basic principle of trust aggregation methods is to integrate multipath trust relationships through weighted average and arithmetic mean approaches [18,22].

In addressing the trust construction issue in SNGDM, Wu et al.[6] proposed a trust propagation operator based on Uni-norm, which overcomes the limitation of traditional trust propagation operators that treat trust and distrust as independent factors. This allows for flexible handling of extreme cases (i.e., 0 or 1) within the range of values of the trust propagation operator. To further study the uncertainty of trust, Tian et al. [32] proposed an interval-valued trust propagation model. Kondo et al. [33] clearly demonstrated the impact of trust on large social systems, providing empirical evidence at the overall level for the positive effects of trust predicted by trust propagation theory on idea sharing, collaboration, and productivity.

Recent research has been undertaken to improve the trust propagation operator to more accurately capture various aspects of trust propagation. Liu et al.[7] proposed a trust propagation method that takes into account the coverage of expert knowledge, providing a theoretical basis for the refined modeling of trust propagation models. Wu et al.[8] developed a dual trust propagation operator, which integrates Einstein's addition and multiplication operators, reflecting both the reduction of trust and the increase of distrust, and extends to the process of distributed trust information propagation. Sun et al. [34] studied the optimal path of trust propagation and proposed a dynamic trust propagation model. Tan et al. [35] studied the problem of long-distance trust propagation and explored the impact of the length of the trust chain on the efficiency of trust propagation. The above work enriches the research in the field of trust propagation and lays a good foundation for exploring SNGDM.

Recently, some trust propagation methods have explored the trust decay phenomenon when trust propagates through different nodes from the trust initiator. Yuan et al. [36] designed a binary tuple linguistic trust propagation operator based on the discount factor to punish longer propagation chains. Ji et al.[9] investigated an exponentially decayed trust propagation operator based on an adjustable decay factor. In fact, trust decay is related to various factors, such as the length of the trust propagation path and the importance of trust nodes, which affect the reliability of the group trust network modeling. Therefore, it is still necessary to systematically consider the influencing factors and coupling mechanisms of trust decay, and to explore the complexity issues of trust propagation to improve the accuracy and reliability of trust modeling in SNGDM.

3.1.3. Trust evolution in social network

Trust relationships within social networks are one of the key factors driving group consensus. These trust relationships reflect the historical interactions of decision-makers, and feedback opinions induced by trust relationships are more readily accepted by the group. Traditional static trust relationships fail to consider the impact of the group interaction on the trust relationship, so it is crucial to characterize the dynamic changes in trust relationships between decision-makers. Teng et al. [37] induced changes in social trust relationships between decision-makers through the gray correlation coefficient between them. Hao et al. [38] proposed a dynamic trust consensus model considering overconfidence. Liu et al. [39] proposed a consensus optimization model based on minimal adjustment of trust relationships to update trust relationships between experts. Xing et al.[10] designed a maximum entropy model to measure the difference in the degree of conflict between decision-makers before and after the modification of the opinion, which is used to update the trust relationships between decision-makers. Yang et al. [40] proposed a link prediction method to describe how decision-makers establish new links with others and designed different rules to explain how decision-makers update their trust relationships with others.

The interaction of opinions between decision-makers leads to dynamic changes in their trust links, and thus the trust network among

[6] https://doi.org/10.1016/j.knosys.2016.01.004.

[7] https://doi.org/10.1016/j.asoc.2020.107005.

[8] https://doi.org/10.1016/j.inffus.2020.10.010.

[9] https://doi.org/10.1109/TFUZZ.2024.3398591

[10] https://doi.org/10.1016/j.inffus.2023.01.004.

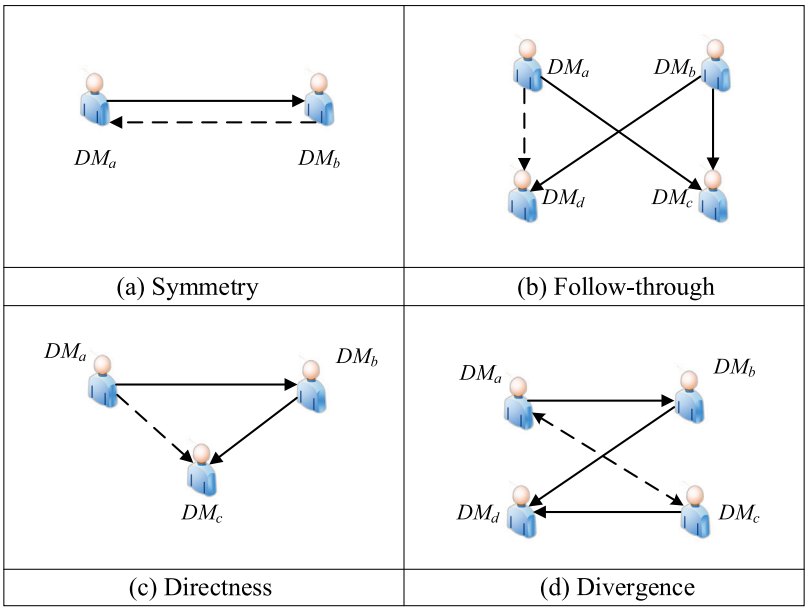


Fig. 7. Characteristics of trust propagation.

groups will evolve dynamically. Establishing the dynamic trust evolution mechanism of the social network helps to clearly demonstrate this process. Zhang et al. [41] proposed a 0–1 integer optimization model with minimal adjustment of trust relationships, leading to changes in the network structure of trust relationships. Wu et al.[11] designed a Hegselmann–Krause-based social network update mechanism to reflect the dynamic changes in the social network structure between experts. Liu et al. [10] designed a consensus feedback strategy that combines opinion adjustment and dynamic trust interaction, where trust relationships between group decision-makers remain unchanged or are enhanced as their opinions are adjusted, and this trust interaction also leads to changes in the social network structure between decision-makers to improve consensus efficiency. Zou et al. [42] proposed a consensus model of trust evolution that considered dynamic limited compromise behavior, indicating that the stronger the trust relationships between subgroups, the higher the compromise threshold of the subgroups and the greater the degree of adjustment of the compromise of the subgroups, which is conducive to efficiently promote consensus achievement.

The work of the aforementioned researchers has enriched the study of the impact of dynamic trust relationships on consensus building. Trust evolution in social network is a complex process involving multiple factors and dimensions, and the evolution mechanisms and their reliability still need to be further explored.

3.2. Internal structure of SNGDM

Clarifying the internal structure of SNGDM is the basis for conducting relevant research. SNGDM uses consensus measurement to determine the state of consensus of the group. In a non-consensus state, community clustering is required to reduce the decision making dimension and complexity. In addition, exploring the power structure relationship of subgroups to propose a reasonable weight-allocation method is also one of the key issues in SNGDM. The following is a review from three aspects: group consensus measurement in social network, group power structure of social network, and community structure of social network.

3.2.1. Group consensus measurement in social network

In SNGDM, measuring the level of consensus among DMs to ensure an acceptable level of group consensus is a necessary prerequisite to obtain reliable decision results [43]. The traditional consensus measurement method aims to measure the level of agreement of the opinions of members of a social network. Gou et al. [44] believe that a two-layer hierarchical clustering method based on pairwise measurement of linguistic distance can achieve high consistency in subgroup decision making. By directly calculating the distance between the aggregated subgroup preferences and the group preferences, the group's consensus degree is obtained. Liang et al. [45] presented a large-scale group decision making framework mining ordinal–cardinal group consensus. Zhou et al. [46] explores trust network-based group decision making with ordinal consensus measure.

The above works treated decision-makers as independent individuals without connections. However, decision-makers often have complex trust relationships, and their opinion can influence each other, thereby affecting the consensus measurement results. Therefore, in addition to consider the similarity of opinions between members, the trust relationships between individuals should also be integrated in the consensus measurement of social network group. In this regard, Dong et al.[12] proposed a group division method based on trust networks that introduces the concept of "leaders" to guide the group to reach a consensus efficiently by managing "leaders". Liu et al. [14] pointed out that individuals with similar opinions but lacking trust relationships may still fail to reach consensus due to relational conflicts. Tian et al. [47] proposed a community discovery algorithm based on interval type-2 fuzzy numbers to find subgroups with close trust relationships, and found that subgroups with mutual trust among members have greater potential for consensus. Wu et al.[13] assumed that decision-makers are more willing to accept suggestions from members they trust, and proposed a feedback mechanism to generate personalized recommendation opinions based on trust networks, thereby enhancing the group's consensus degree.

In general, consensus measurement of social network group by combining opinion similarity and trust degree is considered a relatively reasonable consensus measurement method. More specifically, a

[11] https://doi.org/10.1109/TSMC.2022.3178230.

[12] https://doi.org/10.1016/j.ins.2017.02.052.

[13] https://doi.org/10.1109/TCYB.2021.3076420.

comprehensive group consensus measurement should include multiple factors such as the consensus degree within subgroups, the consensus degree between subgroups, and the trust relationship among group members.

3.2.2. Group power structure of social network

The power structure of a social network describes the organizational system of a group, including both the distribution of power within the group and the influence relationships between different powers within the organization, which directly affects the efficiency of consensus reaching in SNGDM and the final aggregation of decision opinions. Rodríguez et al. [48] proposed a subgroup cohesion measurement method based on restricted equivalence functions to reduce the impact of internal disagreements within subgroups driven by the "majority principle" decision making. Tang et al. [49] proposed a method for determining weights that combines subgroup size with indicators of clustering result effectiveness to investigate SNGDM with non-cooperative behavior.

The above work mainly studies the subgroup empowerment methods from the perspective of subgroup size, but the impact mechanism of weight distribution on the change of group consensus state has not been investigated. Xu et al. [50] noted that the feedback mechanism in group decision making will change the decision-makers' original preferences, thereby affecting the weighting information based on decision-maker preferences. Tan et al. [51] studied the consensus reaching process in dynamic trust environments and found that subgroup weights will be dynamically updated with the update of trust relationships. Chao et al. [52] studied group decision making with non-cooperative behavior and heterogeneous preferences, and promoted group consensus by reducing the weight of non-cooperative decision-makers. These studies indicate that subgroup power management is an important means of optimizing group consensus.

However, existing subgroup power allocation methods mostly rely on explicit performance of subgroups, such as subgroup size and different power structures. It is necessary to systematically examine the effect of power structures on consensus reaching, including implicit power characteristics, so as to achieve the goal of optimizing decision making resources and improve the consensus efficiency of SNGDM through subgroup power management.

3.2.3. Community structure of social network

As the complexity of decision making problems and the scale of decision-makers continue to grow, clustering methods have been widely used to reduce the dimension of group size in group decision making problems. Cluster can divide large groups of decision-makers in social networks into multiple subgroups, which helps to improve the efficiency of subgroup governance and consensus-reaching processes.

(1) **Traditional clustering methods.** Traditional research on subgroup structures assumes that decision-makers are independent of each other, without considering social connections and historical interactions between decision-makers. Wu et al.[14] used the k-means method to group social networks in a fuzzy preference environment and tested the impact of the clustering results on the convergence of consensus models. Tang et al. [53] considered both the levels of similarity within and between subgroups in large-group decision making, proposing an adaptive consensus model based on the fuzzy C-means clustering algorithm. Pan et al. [54] adopted a clustering model based on a dynamic programming algorithm, which can perform iterative clustering based on the similarity of decision-makers' evaluation information. Wu et al. [55] implemented a clustering algorithm for subgroup division based on consensus and consistency measures of decision-makers' preference relationships.

(2) **Trust based clustering methods**. In actual decision making scenarios, the complex trust relationships between decision-makers can have a significant impact on the clustering process and outcomes of group decision making. Therefore, some scholars have begun to introduce trust relationships to address social network group clustering problems. Yu et al. [56] proposed a trust Cop-Kmeans semi-supervised clustering algorithm, which uses trust relationships generated by prior knowledge as clustering constraints. Liu et al. [13] believed that the preference information of decision-makers is sometimes not reliable enough and proposed a trust network clustering model based on social network analysis. Based on the gray clustering algorithm, Lu et al. [57] established a clustering method to reduce the consensus complexity of network group decision making.

(3) **Overlapping communities in social network.** In SNGDM, social network overlapping communities are special community structures that can facilitate opinion interaction to achieve group consensus. The existing research has investigated the impact of intricately overlapped social trust relationships within overlapping communities on the preference modification in CRP of SNGDM from different perspectives. Tang et al. [58] proposed a delegation mechanism-based consensus method with heterogeneous experts and overlapping communities. Ji et al.[15] established a trust-driven consensus model in SNGDM with a decayed trust propagation method among multiple overlapping communities. Wang et al. [59] introduced a CRP that integrates overlapping community detection and ELICIT-based optimization models under bounded confidence. Gai et al. [60] made exploration into the specific mechanisms of different overlapping structures facilitating community interaction and proved that overlapping communities can improve the compromise willingness of experts compared to non-overlapping communities, indicating that overlapping communities can serve as a bridge to facilitate interaction. Overall, scholars have conducted necessary explorations on the role of overlapping communities in subgroup interaction and trust propagation. Overlapping nodes serve as "intermediaries" or "bridges" connecting different communities, analyzing the trust modeling or consensus feedback mechanism under the overlapping structure is worthy of in-depth exploration.

It should be noted that although clustering technology strengthens the homogeneity within subgroups, it also strengthens the heterogeneity outside the subgroups. Therefore, high internal consensus within a subgroup does not necessarily mean that the subgroup is highly consistent with other subgroups, which will lead to low-quality decision making results even under high consensus levels, resulting in the phenomenon of "pseudoconsensus" of decision making. Therefore, a reasonable subgroup clustering structure should consider a variety of factors, and the impact mechanism of each factor on different social network community clustering should be explored, so as to facilitate a subsequent consensus-building process.

3.3. Consensus mechanism of SNGDM

The critical aspect for reaching consensus in SNGDM is that the consensus opinion is a guarantee of the reliability of decision making solutions and is more easily accepted by decision-makers. In social network scenario, the trust relationship between decision-makers can play a promoting and incentivizing role for CRP. In addition, in the process of consensus interaction, decision-makers will exhibit non-cooperative behavior and manipulative behavior that hinder group consensus. Trust-based behavior management has also been widely

[14] https://doi.org/10.1109/TFUZZ.2019.2953602.

[15] https://doi.org/10.1109/TFUZZ.2024.3398591.

studied by scholars. The following is a review from three aspects: feedback mechanism for consensus in social network, SNGDM problems with opinion dynamics, and trust based behavior management in social network.

3.3.1. Feedback mechanism for consensus in social network

(1) **Traditional feedback mechanism.** Existing consensus group decision making interaction mechanisms mainly study how individuals/subgroups adopt group opinions, which is essentially an interaction mechanism where individuals/subgroups continuously converge towards the group. Li et al. [61] pointed out that there are mainly two consensus rules used in group consensus decision making: feedback mechanisms based on recognition-direction rules and feedback mechanisms based on optimization rules. Zhang et al. [7] detected the decision-makers' subnetworks and identified opinion leaders through network segmentation algorithms, and then designed a feedback mechanism based on recognition-direction rules considering the experts' leadership and limited trust. Zha et al. [62] proposed a multistage optimization feedback mechanism based on individuals' bounded confidence, aiming to maximize the group's mutual acceptance and minimize preference adjustment. Cheng et al. [63] analyzed individuals' limited compromise behavior and tolerance behavior, and proposed a minimum cost consensus model with direction constraints considering tolerance and a minimum cost consensus model with threshold-based direction constraints. Du et al. [64] used an extended K-means method to divide the group into different subgroups and enabled a feedback interaction mechanism to improve the consensus level of the subgroups. Guo et al. [65] investigated multidimensional multiround feedback mechanisms with reward and punishment measures for the attainment of group consensus.

(2) **Trust driven feedback mechanism in social network.** In SNGDM scenarios, the social trust relationship between decision-makers becomes a necessary factor in driving consensus interaction and enhancing feedback willingness. Wu et al.[16] constructed a two-level personalized feedback mechanism for consensus issues in SNGDM. Zhang et al.[17] proposed a social trust driven minimum adjustment consensus reaching feedback mechanism considering the willingness of experts to modify their assessments. Li et al. [66] developed a feedback mechanism to adjust experts' opinions based on bounded confidence and social network. In the democratic decision making environment under the social network, Liu et al. [67] considered the personal attributes of the participants and proposed a consensus feedback strategy based on the degree of compromise. Liu et al.[18] proposed a trust-based recommendation mechanism to generate personalized suggestions for experts with inconsistent opinions to enhance the overall consensus level. Sun [68] developed a consensus-reaching process based on herding behavior in the context of social network large-scale group decision-making. In 2024, Jin et al. [69] investigated a personalized bidirectional feedback mechanism that combines cooperation and trust to improve group consensus in SNGDM and Xing et al. [70] proposed a trust incentive rule induced by the dynamic trust relationship that captures the decision-maker's willingness to adapt in order to facilitate the consensus process in an efficient way.

These existing research works have explored the promotion and incentivizing role that trust plays in feedback mechanisms. As an essential indicator in SNGDM, the trust level has a complementary and mutually promoting relationship with the consensus level. Systematically exploring the incentives and promotion effects between trust relationships and consensus interaction is of great significance for the consensus issue in SNGDM.

3.3.2. SNGDM problems with opinion dynamics

In social networks, the transmission and evolution of user opinions along network relationships is the dynamics of opinion. The dynamics of opinion explores the process of forming opinions in a group of agents, which plays a significant role in SNGDM [71]. Dong et al.[19] developed a consensus building process in opinion dynamics based on the concept of leadership in the social network. Li et al. [72] proposed a multiattribute group decision making method with opinion dynamics based on the social trust network. Dong et al. [73] proposed a dynamic model of opinion based on individual and local world opinion to reach consensus on social networks. Li et al. [74] investigated an iterative algorithm of social power and opinion evolution to determine the final consensus opinions in large-scale group decision making. Wu et al. [9] constructed two new mixed opinion dynamics models to more realistically describe the opinion evolution process in social networks.

In addition to the above studies that focus on the opinion dynamics from theoretical perspectives, there are also researchers who aim to understand societal phenomena such as the formation of public opinion, the rise of social movements, and the acceptance of public policies by analyzing how individuals receive and process information through social interactions and how these interactions influence their opinions and decisions. In this field, research can focus on developing new models and tools to help understand and predict group decision making processes within social networks. Wu et al.[20] proposed a new model of evolution of public opinion of the leader–follower on the Suez Canal blockage event. Zhuang et al. [75] developed an emotional evolution classification model for large-scale online public opinion on events such as COVID-19.

However, in the process of the evolution of opinion within social networks, the spread of misinformation and the manipulation of opinions are increasingly drawing attention. Due to the fast pace and wide reach of information dissemination, inaccurate or biased data can quickly spread throughout the network, affecting the beliefs and decisions of a large number of users. In addition, some influential individuals or groups may deliberately disseminate misleading information to manipulate public opinion, thereby creating divisions in society or promoting specific agendas. Therefore, future research will focus on how to effectively curb the spread of misinformation in social networks, investigate the reasons for persistent disagreements among different individuals, develop theoretical models of belief manipulation and indoctrination, and enhance society's resilience against the dissemination of misinformation.

3.3.3. Trust based behavior management in social network

(1) **Manipulation behavior management.** Weight manipulation is the behavior generated by individual during the preference aggregation process, which refers to the intentional alteration of weights or influencing factors by individuals or entities in evaluation process to achieve specific goals or outcomes. Weight manipulation behavior, while potentially beneficial for achieving certain objectives and optimizing resources, can also lead to distorted decision making, unfairness, and trust issues. Consequently, it often requires constraints such as transparency, regulation, and punishment to ensure its use is legal and fair [76]. Yager [77,78] explored individual weight manipulation behavior during the group preference aggregation process, and proposed management mechanisms to counteract such behavior. Dong et al.[21] investigated strategic weight manipulation in multiattribute group decision making problems. Liu et al.[22] studied strategic weight manipulation in the context of group decision

[16] https://doi.org/10.1109/TCYB.2021.3076420.
[17] https://doi.org/10.1109/TFUZZ.2021.3073251.
[18] https://doi.org/10.1016/j.knosys.2016.12.014.
[19] https://doi.org/10.1016/j.ins.2017.02.052.
[20] https://doi.org/10.3390/jmse10122006.
[21] https://doi.org/10.1016/j.omega.2017.02.008.
[22] https://doi.org/10.1109/TSMC.2018.2874942.

making with interval attribute weight information from an optimization perspective and proposed a weight manipulation model under a minimum cost strategy. Sun et al.[23] proposed a minimum cost anti-manipulation feedback model that balances fairness and efficiency to help groups reach consensus.

Furthermore, weight manipulation behavior can undermine trust, leading to increased distrust among users, thereby affecting the quality and effectiveness of decision making. Therefore, it is necessary to study the relationship between weight manipulation and group consensus. In 2021, Dong et al.[24] proposed strategies that combine feedback mechanisms, leader preference adjustment, and trust relationship improvement in group decision making to achieve consensus, warning of the manipulation of trust relationships and their countermeasures. Xiong et al. [79] used trust and preference historical data to identify and address strategic manipulation behavior in large-scale group decision making, and implemented punishment strategies for subgroups exhibiting manipulative behavior by updating weights. Liang et al. [80] proposed an improved identification and management method based on trust relationships to prevent manipulative behavior by decision-makers.

In summary, current research has explored individual weight manipulation issues from different perspectives. It is necessary to identify potential user manipulation behaviors in SNGDM to ensure the authenticity and fairness of the aggregation information. When analyzing such behaviors, the social network relationships between users become an indispensable factor, in-depth research on the trust relationships and collaboration patterns between users can help to detect manipulators.

(2) **Non-cooperative behavior management.** CRP usually involves multiple rounds of interaction and discussion in which decision-makers are required to make concessions on their initial positions and revise their preferences. In this process, decision-makers may refuse to cooperate and be unwilling to make compromises, which is defined as exhibiting non-cooperative behavior. Non-cooperative behavior will hinder opinion interaction and consensus improvement [81]. Existing research has conducted studies on the identification and management of non-cooperative behavior from different perspectives. Sun et al.[25] proposes a framework to manage subgroups' non-cooperative behavior by weight penalty in large-scale group decision making. Tian et al. [82] developed an adaptive consensus framework to support heterogeneous LSGDM, in which a two-stage uninorm-based behavior noncooperative management method is designed to generate personalized weight feedback to each DM and subcluster.

In the context of social networks, the emergence of non-coopera tive behavior may be related to insufficient trust between individuals. Therefore, it is worth exploring the addressing of non-cooperative behavior from the perspective of regulating trust relationships. Li et al. [83] developed a non-cooperative behavior detection and management method for CRP based on opinion evolution under the WeChat-like interaction network context. Yang et al.[26] defined three new categories of non-cooperative behaviors, bribery behavior, passive participation behavior, and potential conflict behavior, and proposed a confidence and trust-based CRP to address these behaviors. Yuan et al. [84] designed a budget-constrained consensus method with minimum group conflict based on social network analysis to address non-cooperative behaviors.

In a word, accurately detection and reasonably addressing of non-cooperative behavior in SNGDM is a necessary prerequisite to obtain decision results that can reach acceptable consensus level. With the in-depth development of SNGDM, studying the transformation between cooperative and non-cooperative behavior from the perspective of enhancing trust relationships is a direction worth considering.

4. Future directions: key SNGDM methodologies

This Section explores the challenges and future directions of SNGDM development from one dimension, key SNGDM technologies. This will be complemented by Section 5, from a general AI perspective and a particular LLMs one, as shown in Fig. 8.

4.1. Trust incentives in SNGDM

Incentive behavior often stems from the individual's internal motivation and external needs. These motivations and needs are seen in psychology as the key factors that drive people's actions. Therefore, a reasonable incentive mechanism can stimulate people's enthusiasm and creativity, making them more actively participate in the decision making process, so as to improve the efficiency of decision making. For example, Yu et al. [85,86] proposed a dynamic incentive variable weight model to improve decision making efficiency. Guo et al. [65] proposed a minimum-cost consensus model based on a reward and punishment mechanism. However, these trust incentives do not take into account the incentive of opinion among decision-makers. Furthermore, the incentive mechanism mentioned above is not considered in the decision making of social network groups. In order to do so, Xing et al. [87] proposed a trust incentive rule induced by a dynamic trust relationship that captures the decision-maker's willingness to adapt in order to facilitate the consensus process in an efficient way, which does not take into account the decision-makers' personality characteristics.

In the realistic decision case, decision-makers may have different personality characteristics, such as conservative, neutral, and aggressive types, which have some influence on both the evolution of opinions and the evolution of trust. Therefore, one issue is that it is necessary to design a personalized trust incentive mechanism with personality characteristics, including the opinion incentive and the trust incentive, to explore the impact on the adjustment of the view opinion and the dynamic change of trust of decision-makers. Another issue is that the trust relationship among decision-makers is also changed through the following trust incentive evolution model. This indicates that the incentive-driven minimum adjustment cost consensus model and the trust incentive evolution model are complementary to each other. Then, trust should be considered as a continuous evolving process, and therefore it is necessary to study the spontaneous process.

4.2. Trust game in SNGDM

Trust Game (TG) is a classic game theory for studying human trust behavior [88], and it can observe cooperative behavior between trustors and trustees in group interactions through the trust game [89]. The combination of trust games and SNGDM provides a tool for understanding and analyzing trust dynamics in social interactions. It involves establishing effective trust mechanisms in the social network to promote consensus cooperation and improve the quality of decision making, which is a complex and challenging research direction.

On the one hand, researchers can use trust games to simulate and analyze how individuals establish trust relationships and adjust their trust levels through repeated interactions and information feedback on social networks. This helps us better understand how people build trust based on past experiences and mutual signals in the absence of centralized authority. However, trust games can help identify and

[23] https://doi.org/10.1007/s10462-022-10361-8.
[24] https://doi.org/10.1109/TSMC.2019.2961752.
[25] https://doi.org/10.1109/TFUZZ.2023.3260820.
[26] https://doi.org/10.1016/j.inffus.2023.101820.

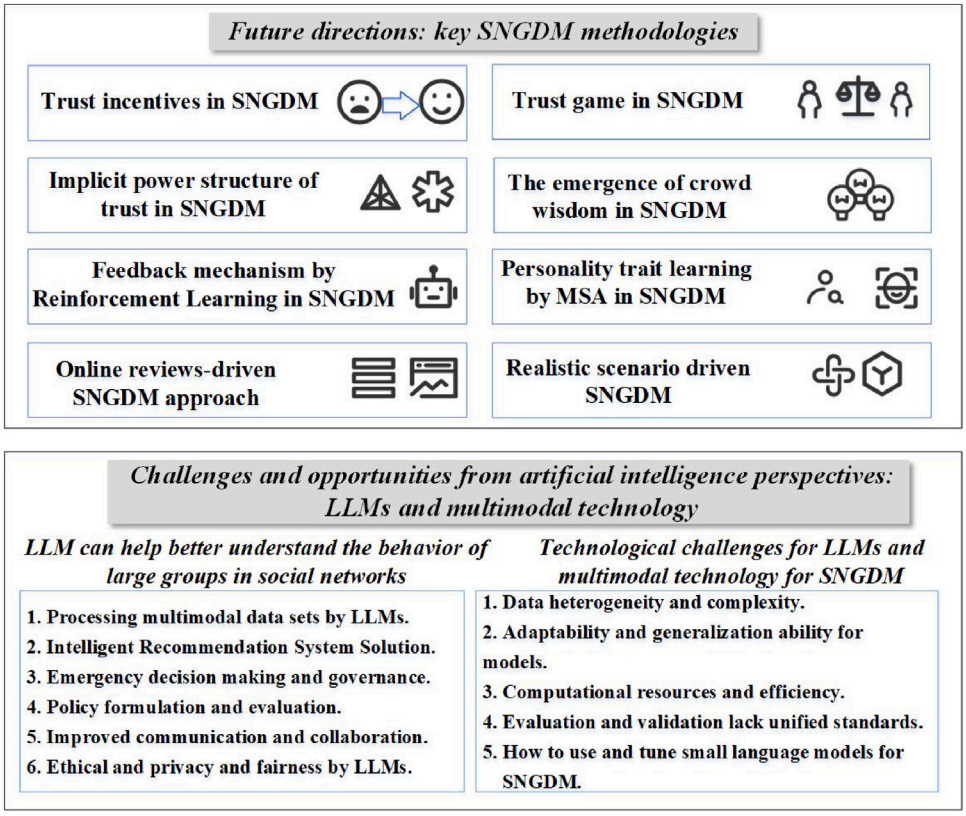


Fig. 8. Overview of challenges and future directions on SNGDM (Sections 4 and 5).

strengthen key factors that promote consensus cooperation. By analyzing the interaction patterns of individuals within a group, mechanisms that promote consensus formation can be designed, such as incentive mechanisms, feedback mechanisms, and evaluation systems [90]. In particular, researchers can also focus on the strategic evolution of repeated trust games [91] between individuals on social networks and how this evolution affects the formation of group consensus. Future research is expected to use trust game theory to delve into the construction of trust mechanisms and the evolution of trust dynamics within social networks, to foster the formation of group consensus, and to enhance the quality and efficiency of SNGDM.

4.3. The implicit power structure of trust in SNGDM

Trust relationships exert an implicit power by shaping the psychological and behavioral dynamics of decision-makers within interpersonal networks, allowing trusted individuals to influence others' actions without formal authority [92]. This implicit power constitutes a novel research direction in SNGDM, focusing on exploring the interplay of influence and being influenced within trust networks, and elucidating their mechanisms in facilitating consensus formation. Implicit power derived from trust relationships endows trusted decision-makers with enhanced influence and persuasive capabilities, enabling them to assume leadership roles in facilitating the convergence of perspectives and the formation of consensus. Furthermore, the implicit power derived from trust can reshape decision making structures, thereby altering the engagement and procedures of decision-makers. This has led to mechanisms that optimize consensus building processes, such as the allocation of trust-based decision making authority and adaptive feedback mechanisms tailored to individual levels of trust capital [53].

It should be noted that research suggests that decision-makers can strategically manipulate trust relationships to achieve desired outcomes, demonstrating that the implicit power of trust influences the allocation of decision making resources, thereby stimulating research into managing trust power. Although the influence of implicit power derived from trust relationships has been acknowledged, research on its dynamic evolution, which includes mechanisms of generation, transfer, and dissipation, as well as the overarching influence of these power structures on the formation of group consensus, remains underdeveloped. Addressing these research gaps has significant potential to optimize collaborative mechanisms within SNGDM.

4.4. The emergence of crowd wisdom in SNGDM

The emergence of crowd wisdom in SNGDM refers to the phenomenon of collective wisdom that generated through interaction, information exchange, and integration between individuals exceeds the sum of individual wisdom when a large number of decision-makers participate in group decision making in the social network environment.[27] When users' various thoughts collide and merge on social networks, new viewpoints and creative ideas are generated. This crowd wisdom transcends the limitations of individuals and provides more possibilities for solving complex problems. For example, when faced with public emergencies, users on social networks can quickly spread information, provide assistance and advice, and jointly contribute to the response to the crisis.

Existing research has explored under what conditions interactive networks within a group can adapt to the dispersion of individual abilities and promote collective wisdom, and discovered that network plasticity and feedback provide fundamental mechanisms for both improving individual judgments and inducing collective wisdom of the network [93]. In future research, the impact of the topological structure of social networks on crowd wisdom can be analyzed, exploring how structural features such as connection patterns, node degree distribution, and clustering coefficient affect the spread of information and

[27] https://doi.org/10.48550/arXiv.2403.15587.

the interaction of opinions. In addition, the dynamic evolution of group decision making can be further examined to explore the dynamic characteristics of group decision making and the role of consensus formation and disagreement resolution mechanisms in the formation of crowd wisdom. Finally, it needs to investigate the impact of group diversity on the emergence of crowd wisdom and how the diversity of group members' backgrounds, professions, and experiences affects information exchange and group collaboration.

4.5. Feedback mechanism by reinforcement learning in SNGDM

Reinforcement Learning (RL) is a machine learning approach that enables agents to learn optimal decision making strategies through iterative interactions with their environment. RL's adaptive mechanisms address several critical elements of consensus formation in SNGDM [94]. One of these is learning through feedback, where agents continually refine their strategies based on interactions with other group members, using feedback to guide adjustments in decision processes. Another is that RL can capture dynamic state updates, allowing agents to adapt to changes in others strategies within the network, not only to their own decisions. Furthermore, RL supports effective dissemination of information, empowering agents to influence others' decisions by optimizing the timing and content of shared information, promoting consensus formation.[28] Recent studies highlight RL's effectiveness in modeling repeated interactions that align individual preferences with group goals, emphasizing its potential to strengthen feedback structures essential for consensus in SNGDM [95].

Despite these promising applications, RL in SNGDM faces several challenges. Firstly, the dynamic nature of social networks demands algorithms that can adapt to changing structures, ensuring that consensus mechanisms remain effective over time. Secondly, information asymmetry between agents, due to differing access to information, poses obstacles to inclusive decision making. Thirdly, achieving convergence in multi-agent settings can also be complex, as aligning reward mechanisms is essential for reaching consensus efficiently. Finally, individual biases complicate consensus, as varying preferences can hinder alignment, requiring careful balance of influence between agents.

Future research may focus on developing adaptive RL algorithms capable of responding to evolving network conditions and on promoting diversity and inclusivity in group decision making to enable broader consensus. Enhancing algorithm transparency and interpretability could further foster trust in RL-driven consensus processes, reinforcing its role as an essential tool for advancing cooperative interactions and effective decision making in SNGDM.

4.6. Personality trait learning by MSA in SNGDM

In SNGDM, the analysis of individual personality traits is beneficial for optimizing the decision making process and improving group collaboration. On the one hand, personality traits will affect decision-makers' risk preferences and conflict resolution behaviors. For example, people with high agreeable traits may be more inclined to compromise and cooperate, while those with high competitive traits are more likely to adhere to their original positions and be reluctant to make concessions. Analyzing personality traits can better predict individual behaviors to build corresponding strategies and mechanisms. On the other hand, analyzing personality traits can facilitate the construction and propagation of trust relationships. Different personality traits correspond to different trust building patterns, and individuals with specific personality traits can serve as intermediaries and bridges for trust propagation. For example, extroverted individuals play a crucial role in trust propagation due to their extensive social networks and positive communication styles.

In the research of personality classification and prediction algorithms, the most widely used machine learning algorithms include Random Forest (RF) [96], K-Nearest Neighbor (KNN) [97], Convolutional Neural Network (CNN) [98], etc. In recent years, multimodal data has become increasingly prevalent in various downstream applications. Multimodal sentiment analysis (MSA) [99], as a technology capable of extracting human emotions and viewpoints from language, acoustic, and visual data streams, has attracted extensive attention in academia and industry. MSA can understand the emotional tendencies and opinions of users by analyzing their text, voice, and video content on social media, which is conducive to optimizing content creation. Therefore, it is helpful to learn the personality characteristics of decision-makers from multiple perspectives in SNGDM.

4.7. Online reviews-driven SNGDM approach

The rapid development of social networks has made users or consumers accustomed to posting reviews of products or services on social networks. These massive evaluation data contain consumer preferences, emotional tendencies, requirements, and other information, which are of great significance for consumer behavior, business decision making, and online platforms.[29] Current research mainly extracts evaluation attributes and user requirements from the obtained online review data. For example, Wu's laboratory utilized the pre-trained language model BERT to generate sentence embedding based on online reviews, and then extracted attributes affecting cruise passenger satisfaction through cluster analysis in SNGDM.[30]

In fact, users on social media platforms can correspond to decision-makers in SNGDM. Existing SNGDM research mainly constructs social networks through questionnaires or directly pre-sets social networks . In future research, it is worth exploring to extract users' social relationships from social media platforms to construct real social networks, and explore how the dynamic changes in trust relationships affect the group decision making process. In addition, it is also worth considering to extract users' evaluation data on products or services from online reviews as the opinion input of SNGDM. Furthermore, analyzing users' personal preferences, historical decision records, etc. on the platform can help to infer their risk preferences, personality and behavioral traits, which will also affect the interaction process and decision making results of SNGDM. Finally, future research can focus on the application of the combination of online reviews and SNGDM in specific fields (such as urban planning, business management, online education, etc.), exploring how to transform theoretical research into practical decision-support tools.

4.8. Realistic scenario driven SNGDM

Scenario-driven SNGDM explores how real-world contexts influence collective decision making within social networks. This field recognizes that group decisions are shaped by both social interaction and specific situations. A seminal study by Allcott [100] in the energy sector demonstrated how social comparison within neighborhood networks could reduce household energy consumption, highlighting the power of social influence in driving behavioral change. In the health sector, Wang et al. [101] revealed how geographical and social proximity affect physicians' adoption of online health communities with their influence varying based on local competition levels. The study of scenario-driven group decision making in social networks provides an important perspective for understanding complex social interactions, which can help organizations and individuals make more informed decisions in changing environments.

[28] https://doi.org/10.1007/978-981-99-8132-8_13.

[29] https://doi.org/10.1016/j.engappai.2024.108509.

[30] https://doi.org/10.1016/j.ins.2024.120801.

Future research on scenario-driven SNGDM could further explore how technology affects group decision making and how the decision making process can be optimized to address emerging challenges. It should focus on following key directions. First, exploring social network incentive mechanisms to encourage the formation of close-knit groups and promote positive behavioral changes through targeted rewards, such as in smart communities' health initiatives. Second, we apply concepts of social physics to study the interaction between social structure and individual behavior, particularly in optimizing community engagement and decision making processes. Third, we investigate the role of social learning in shaping group interactions and decision making processes in various fields. Furthermore, empirical studies that test theoretical SNGDM models with real data, develop AI-enhanced predictive models, and examine cross-cultural differences will provide a scientific basis for effective social intervention measures and sustainable community development.

5. Challenges and opportunities for SNGDM from artificial intelligence perspectives: LLMs and multimodal technology

AI is a broad field that involves research and development of technical disciplines for simulating and extending human intelligence. LLMs focuses on natural language processing, that is, completing language-related tasks such as generation and understanding through a large amount of text training; while multimodal technology emphasizes the processing and integration of different types of data modalities to achieve cross-modal tasks. With the support of AI, multimodal technologies and LLMs, the landscape of SNGDM is poised to undergo a transformative evolution. These advances promise to harness vast datasets and diverse forms of input, such as text, images, voice, and even behavioral data, providing a richer, more nuanced understanding of group dynamics. LLMs, with their ability to process and generate human-like text, can facilitate more effective communication and interpretation of group sentiments, enabling more informed and equitable decision making processes. However, this evolution is not without challenges. Ensuring the ethical use of these technologies, addressing biases embedded in AI models, and managing the privacy and security of personal data are critical hurdles that researchers and practitioners must navigate. Furthermore, the integration of these technologies into existing systems will require robust frameworks and interdisciplinary collaboration, blending insights from computer science, sociology, psychology, and ethics.

As we enter this new era, the potential for innovation in the decision making of social network groups is vast, yet it demands a conscientious approach to mitigate risks and leverage AI's capabilities responsibly. Along next two subsections we analyze challenges for a double perspective, a better understanding the behavior of large groups within social networks and technological goals.

5.1. LLMs can assist organizations in better understanding the behavior of large groups within social networks

A significant challenge, particularly with large opinion datasets, is how to effectively integrate and analyze the opinions of large social network databases to support decision-making. LLMs such as ChatGPT can help researchers construct more complex and accurate analytical models to predict trends in social networks, user behavior patterns, and group dynamics [102].

With strong language comprehension and text processing capabilities, as well as efficient information extraction and integration capabilities, LLMs can assist organizations in better understanding the behavior of large groups within social networks. This can lead to more informed decisions on marketing, public safety, and social network management. Specifically, it includes the following aspects:

(1) Public opinion monitoring and governance solutions based on multimodal technology and SNGDM. Using artificial intelligence and data processing technology to process and analyze user behavior big data on social networks is a promising direction for SNGDM in the future. This includes identifying user interaction patterns, performing influence analysis, and studying the dynamics of opinions. In addition, users not only express their emotions through text, but also frequently share images or videos to convey feelings on social media platforms. And user emotional states can be more accurately identified and analyzed by integrating natural language processing (NLP) and computer vision (CV) technologies. Real-time analysis of large volumes of multimodal data enables the prediction of emotional trends for specific events or topics, which is crucial for brand monitoring and public policy formulation.

(2) Intelligent Recommendation System Solution based on LLMs and SNGDM. LLMs can generate precise recommendation content analyzing user behaviors and preferences on social networks. For example, relevant news, articles, or products are recommended based on users' browsing history, likes, and comments. Based on the user's social network graph, graph neural networks, and other large models are used to predict new friends or potential partners that the user may be interested in, thereby enhancing the connectivity of the social network.

(3) SNGDM-assisted emergency decision making and governance solutions. In the context of natural disasters or public health crises, multimodal data analysis can assist in rapid assessment of the situation of affected areas. For instance, by analyzing satellite images and real-time reports on social media, one can more accurately determine the extent of the disaster's impact. Additionally, real-time analysis based on multimodal data can optimize the allocation of rescue resources. For example, after an earthquake occurs, analyzing images and videos from the affected area can help identify which regions are in urgent need of assistance.

(4) SNGDM-assisted policy formulation and evaluation. SNGDM can be used to simulate policy effectiveness, assessing the potential impacts of different policy measures to support multimodal data. For example, by simulating various traffic control schemes, predict their effects on urban traffic flow. Collect multimodal feedback information after the implementation of policies, continuously adjust, and refine policy measures to form a closed-loop management. For example, by analyzing public discussions and evaluations of new policies, adjust policy directions quickly.

(5) Improved communication and collaboration. By processing and synthesizing large amounts of information, large groups, LLMs can enhance the clarity and effectiveness of communication among decision-makers. It can improve communication to support collaborative efforts and consensus building.

(6) Ethical and privacy and fairness considerations. This is an essential responsibility in the current social networks scenario, privacy and ethical issues for decisions. Incorporating robust mechanisms to ensure ethical use of data opinions and protect user privacy is essential, thus fostering trust and compliance with regulatory standards. In addition, LLMs can help identify and address biases within opinions and decision making processes, promoting fairness and equity in outcomes.

In summary, LLMs and multimodal technology provide new perspectives and methods for addressing complex social issues in traditional SNGDM.

5.2. Technological challenges for LLMs and multimodal technology for SNGDM

As technology continues to advance and applications become more sophisticated, this field will unveil an ever-expanding array of possibilities and potential. However, leveraging LLMs and multimodal technologies for SNGDM presents several technological challenges. Below are five key areas that exemplify these challenges:

(1) Data heterogeneity and complexity. The forms of data in social networks are diverse, including text, images, videos, etc. How

to effectively integrate these different modalities of data, extract useful information, and conduct unified analysis is a major challenge. The semantic information in multimodal data is complex and diverse, requiring advanced natural language processing and computer vision technologies for accurate understanding and interpretation.

(2) Adaptability and Generalization Ability for Models. Different social networking platforms may have significant differences in user behavior and content style, requiring models to possess good cross-domain adaptability. The social networking environment is highly dynamic, with constantly changing user behaviors and hot topics, necessitating that models can update and adapt to these changes in real time.

(3) Computational Resources and Efficiency. Social network data is massive, and processing and analyzing this data requires high-performance computing resources. For some application scenarios, such as public sentiment monitoring and crisis management, the system needs to be able to respond quickly and provide real-time analysis results.

(4) Evaluation and validation lack unified standards. It should be noted that the black box nature of LLMs makes its analysis process opaque [103,104], and the training data of different models may be biased, which in turn affects the analysis results. Currently, there are no unified standards for evaluating the effectiveness of the integration of multimodal technology and SNGDM. Due to the complexity of the social network environment, it is challenging to design experiments that can comprehensively assess model performance. Therefore, when integrating LLMs to SNGDM, it is necessary to combine other methods for comprehensive judgment. Future research may concentrate on enhancing the accuracy, efficiency, and interpretability of LLMs.

(5) How to use and tune small language models for SNGDM. The emergence of Small Language Models (SLMs) [105] is revolutionizing the AI landscape by making advanced language processing capabilities more accessible and efficient. SLM compact models offer a balance between performance and resource consumption, enabling deployment on a wider range of devices, including those with limited computational power. This democratization of technology means that more businesses, researchers, and developers can integrate advanced language understanding and generation into their applications, fostering innovation across various fields. Furthermore, the reduced footprint of these models allows for quicker training and fine-tuning, facilitating more rapid iteration and customization to meet specific needs. As a result, SLMs are playing a crucial role in expanding the reach and impact of AI technologies, driving progress and opening new opportunities in countless domains. Utilizing SLMS for SNGDM poses significant challenges. They might struggle with capturing nuanced social interactions and dynamics, leading to potentially less accurate or comprehensive insights. Additionally, ensuring the robustness and adaptability of SLMs in rapidly changing social contexts remains a critical hurdle to overcome but also an advantage to tune an SLMs design for decision making processes.

6. Conclusions and lessons learnt

SNGDM models play a significant role in the integration society of data and intelligence. It facilitates the aggregation of collective wisdom, and through reasonable model design, it orderly integrates diverse information and perspectives, enhancing the scientific nature and rationality of decision-making and reducing blindness and subjectivity. It can accurately analyze group structures and dynamics, identify key nodes and influential factors, and optimize decision-making processes and efficiency. Moreover, it provides new approaches for addressing complex real-world issues, aiding scientific decision-making in business management, political governance, and social planning, enhancing decision-making transparency and democracy, improving plan acceptance and implementation effects, and promoting efficient resource allocation and positive social development.

Therefore, this article reviews and analyzes the state-of-the-art of SNGDM methods from different dimensions. Firstly, it provides a clear description of the SNGDM framework by showing its characteristics. Secondly, it reviews trust relationship analysis methods used in SNGDM approaches to reflect the mutual influence between decision-makers. Thirdly, it analyzes the internal structure of SNGDM problems under a classification of key elements: social network group consensus measurement, community structure clustering, group power structures analysis, as well as trust models and feedback mechanisms for building consensus. As an important goal, it identifies challenges and future research directions in SNGDM studies from a double look, SNGDM methodologies and opportunities for SNGDM from AI perspectives, LLMs and multimodal technology.

The insights and lessons learned from the contributions of this article are as follows:

- Despite increasing attention and depth of research on SNGDM over the past decade, there remains to date a lack of formal definition and characterization of this decision framework.
- SNGDM is inherently complex and uncertain. In addition, different trust relationships exist within social networks, indicating the significant impact of considering trust relationships in SNGDM problems. Various trust representation methods, trust propagation mechanisms, and social network structures all influence the final decision making outcomes.
- LLMs, multimodal fusion technology, and AI technologies are among the main directions for the future SNGDM. This technology contributes to advancing decision science from the era of theoretical models to an intelligent data-opinions-driven era of SNGDM.

This article aims to better assist researchers in understanding SNGDM problems. The goal is developing more new models, placing greater emphasis on the intrinsic mechanisms of trust formation, and exploring the potential applications of next-generation information technology in SNGDM solutions.

As a last reflection, in this new AI scenario, it is worth considering how and for what scenarios (under regulations) to design autonomous AI agents for decision making and consensus. These agents must be equipped with advanced algorithms capable of understanding and processing complex social dynamics and varying opinions within a group. Their design should emphasize transparency, ensuring that their decision making processes are understandable and justifiable to human stakeholders. Moreover, incorporating ethical guidelines and bias mitigation strategies is crucial to promote fair and equitable outcomes. Additionally, these AI agents should be adaptable, able to learn and evolve with changing group dynamics and contexts, ensuring their continued relevance and effectiveness. By thoughtfully integrating these elements, we can create AI agents that not only support but enhance collaborative decision making processes in social networks.

CRediT authorship contribution statement

Mingshuo Cao: Writing – original draft, Formal analysis, Data curation. **Tiantian Gai:** Writing – original draft, Formal analysis, Conceptualization. **Jian Wu:** Writing – original draft, Supervision, Investigation, Funding acquisition, Conceptualization. **Francisco Chiclana:** Writing – review & editing, Validation. **Zhen Zhang:** Writing – review & editing, Visualization, Conceptualization. **Yucheng Dong:** Writing – review & editing, Validation, Conceptualization. **Enrique Herrera-Viedma:** Writing – review & editing, Validation. **Francisco Herrera:** Writing – review & editing, Visualization, Validation, Conceptualization.

Declaration of competing interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled.

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Data availability

No data was used for the research described in the article.

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